ASSESSMENT ACTIVITY COURSE DETAILS

Activity Marks: 5 **%age Marks out of 100:** 5 %	**ASSESSMENT ACTIVITY** ASSIGNMENT 2	**Obtained Marks:** **%age Marks Obtained:**
Student Name: **Student ID:**	**TEACHER:** Tabassum Zaman	**Activity Date:** 04-10-2025 **Deadline Submit:** 15-11-2025
Problem Complexity (CP/CA): CP3/CA2 **Knowledge Area / Sub Area:** SP **Seoul Accord Attribute (GA):** GA-1 *(See NCEAC Curriculum / Seoul Accord for details)*	**COURSE CODE & TITLE:** SES-311 Simulation & Modeling	**SEMESTER:** 5th semester **OFFERED IN**: Spring 2025 **FOR THE BATCH**: Fall 2023

OBE MAPPING AND LEARNING DOMAINS

Course Learning Outcome (CLO) Covered	Graduate Attributes (GA) Covered	Learning Taxonomy Domain and Level
CLO-2	GA-2 (Knowledge for solving Computing Problems)	C3 Apply
Covers introduction, basic simulation concepts, and model types conceptual and foundational understanding.	*Explain the basic concepts of simulation and modeling, including types, purposes, advantages, disadvantages, and components of simulation models.*	*Cognitive Domain Level-2 Attribute: Understanding what the facts mean.*
Sustainable Development Goals (SDGs): Awareness of **SDG9** (Industry Infrastructure) *(See UN website for further details)*	**SDGs Related Target #:** Awareness of SDG Target **9** *(See UN website for further details)*	**Professional Competence Profile:** Engaged in related topics *Graduates of computer science programs should have fundamental competency in the areas described by the Body of Knowledge (see Chapter 4, CS Curricula by ACM), particularly the core topics contained there.*

ASSESSMENT ACTIVITY INSTRUCTIONS

1- This assignment activity has total **5 Marks** (5% Weightage) and individual marks of each question is also mentioned.
2- The due date of assignment submission for checking is **04-Nov-2025.**
3- The due date of assignment for uploading on CMS is **11-Nov-2025.**
4- This assignment covers the **CLO-2** of the course as mentioned in course profile.
5- This assignment is mapped with **PLO-2/GA-2** (Knowledge for solving Computing Problems) of the program.
6- This assignment will be evaluated for learning of **C3** (Level-3 of Cognitive domain).
7- This assignment covers awareness of **SDG9** (Industry Infrastructure)
8- This assigned based on fundamental computer competency in the areas described by the Body of Knowledge in ACM
9- Assignment must be submitted in **neat and clean** Hand-written form (Preferably A4 size paper).
10- Assignments must be submitted **with these cover pages** after filling the student Name and ID.
11- Assignment must be with **proper numbering** of Pages, Figure, Tables and Equation if available.
12- This assignment should be solved by students themselves & **mention references** as per standard format (If Applicable).
13- This assignment will be rejected in case of exceeded allowed **limit of Similarity or found Plagiarized** (If Applicable).
14- This assignment length is restricted to (1) Number of **Pages** _N/A_ and (2) Number of **Words** _N/A_ (If Applicable).
15- Assignment must be submitted with **proper file cover** before due date mentioned on CMS.
16- **Hardcopy** of Checked Assignment must be kept **in your record** till final result or corrections in obtained marks.
17- Softcopy of Assignment must be **uploaded on CMS** within the due date otherwise marks will not be entered on CMS.
18- Make sure to **save the scanned file name** as
(Short Course Name) + (Activity #) + (Semester S/F-XX) + (Student ID) => **Engg Eco Assign#1 S-24 (3254-2021)**

ASSESSMENT TASK

Activity Marks: 5 **%age Marks out of 100:** 5 %	**ASSESSMENT ACTIVITY** ASSIGNMENT 2	**Obtained Marks:** **%age Marks Obtained:**
Student Name: **Student ID:**	**TEACHER:** Tabassum Zaman	**Activity Date:** 04-11-2025 **Deadline Submit:** 15-11-2025
Problem Complexity (CP/CA): CP3/CA2 **Knowledge Area / Sub Area:** SP **Seoul Accord Attribute (GA):** GA-2 *(See NCEAC Curriculum / Seoul Accord for details*	**COURSE CODE & TITLE:** SES-311 Simulation & Modeling	**SEMESTER:** 5th semester **OFFERED IN:** Spring 2025 **FOR THE BATCH:** Fall 2023
OBE Attainment (Course): Course Learning Outcomes: CLO-2	**OBE Attainment (Program):** Program Learning Outcome (**PLO**) **PLO-2/GA-2:** (Knowledge for solving Computing Problems)	**OBE Attainment Taxonomy):** **Domain:** Cognitive **Level:** C3 Apply
Sustainable Development Goals (SDGs): Overview & Awareness 9	**SDGs Related Target #:** Overview & Awareness 9	**Professional Competence Profile:** fundamental computer competency in the areas described by the Body of Knowledge in ACM

TASK: Carefully Read all Assessment Instructions on main page and answer all the following questions. Each Question carry 10 Marks and Total 5 Marks (overall 5% Weightage in Final Result)	**OBE Attainment**	**Other Related Mapping**	**Obtained Marks**
Question NO 01: Analyze the three steps of simulate a model and also explain them. Compare the discrete and stochastic model and also give an example. **Question NO 02:** Draw a pareto chart and imagine you're a customer service manager at a company and notice that complaints have increased. You want to address the most critical problems first to improve customer satisfaction. You categorize the complaints as follows: • **Product quality issues** (45 complaints) • **Delayed delivery** (30 complaints) • **Poor customer support** (20 complaints) • **Billing errors** (15 complaints) • **Website navigation issues** (10 complaints) •	C3 CLO-2 GA-2	2 marks 2 marks CP3/CA2 SP GA-2 1 marks	2.5 MARK EACH Total 5 Marks

TEXT BOOK & REFERENCE:

1. "Simulation Modeling and Analysis" by Averill M. Law and W. David Kelton 5 TH Edition
2. "Discrete-Event System Simulation" by Jerry Banks, John S. Carson II, Barry L. Nelson, and David M. Nicol 6 th Edition

Examiner Signature: **Marks Obtained:**